Model Training Report

AI-Based Waste Detection and Classification System

1. YOLOv8L

Model Explanation

YOLOv8L is a CNN-based, anchor-free single-stage detector: a CSPDarknet-style backbone extracts multi-scale features, a PAN/FPN neck fuses them, and a decoupled head predicts class and box offsets per grid cell directly, with no separate region-proposal stage. This run overrides Ultralytics' default optimizer with a 2-group differential learning-rate schedule: the first 5 backbone layers (generic edge/texture filters) are frozen outright, the remaining backbone/neck layers train at a low learning rate (3.5x10-4), and the detection head - which must learn this project's 6 classes from scratch - trains at a high learning rate (7.0x10-3), both annealed on a cosine schedule.

Dataset

Dataset is taken from the project's main conveyor/pour capture set - 7,389 training images and 1,847 validation images, hand-annotated in YOLO format across all 6 material classes.

Training Parameters

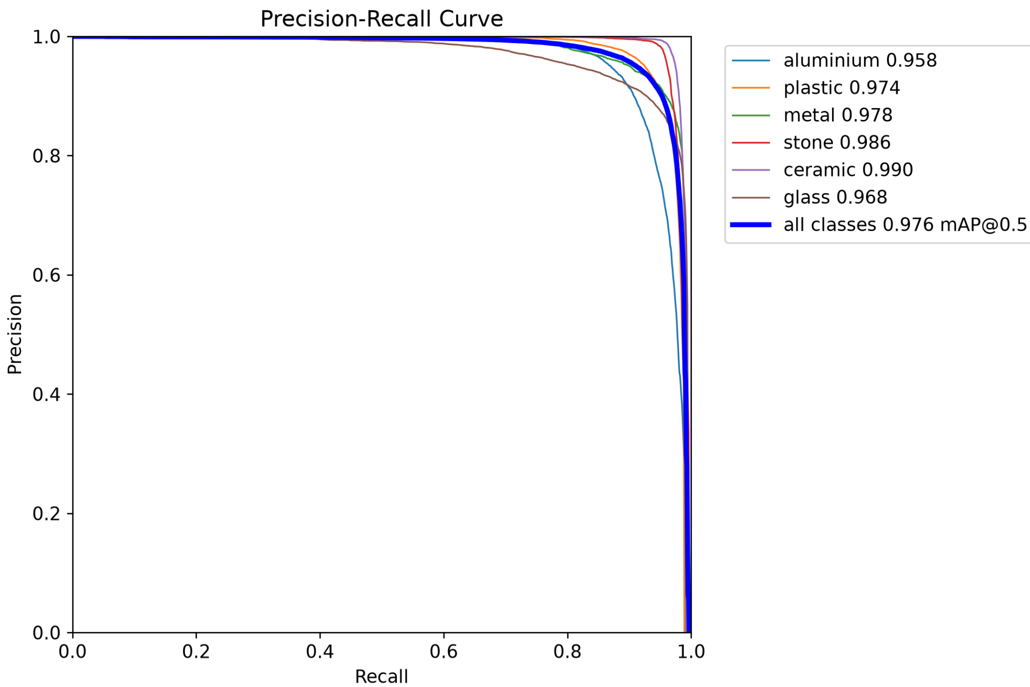
Base model	YOLOv8L
Classes	6 - aluminium, plastic, metal, stone, ceramic, glass
Training images	7,389
Validation images	1,847
Batch size	8
Training Image Input size	640px
Epochs trained	100
Training time	2h 19m (100 epochs)
Precision	0.945
Recall	0.932
mAP50	0.975
mAP50-95	0.935

Average model Inference time is : 5.82 ms (171.8 FPS)

Average end to end latency : 87.61 ms

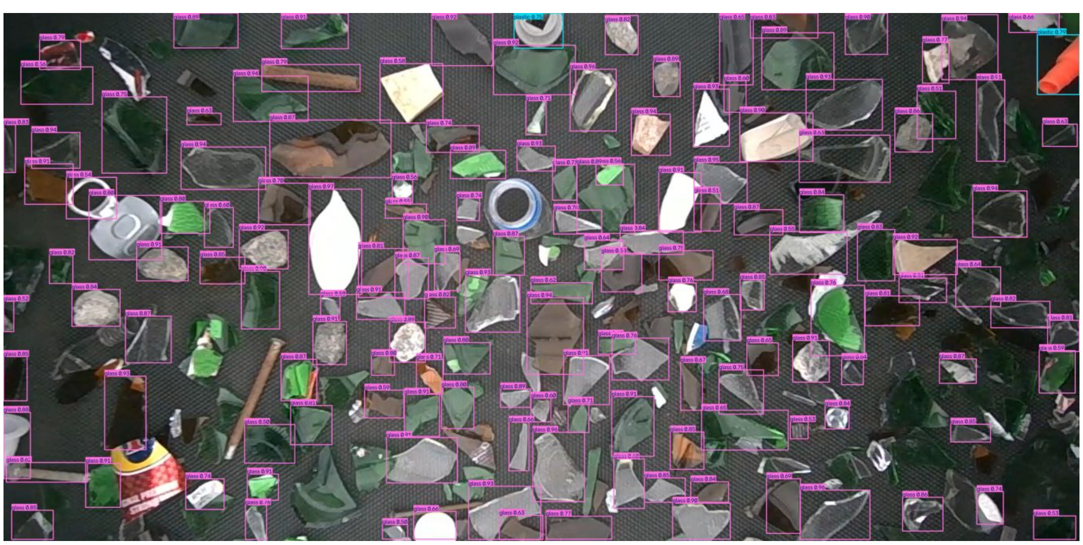
(Measured on an RTX 5080, 3622x1779 test frame, 5 warmup + 30 timed passes. End-to-end includes reading the frame and drawing the annotated output, not just the model's forward pass.)

Model curve

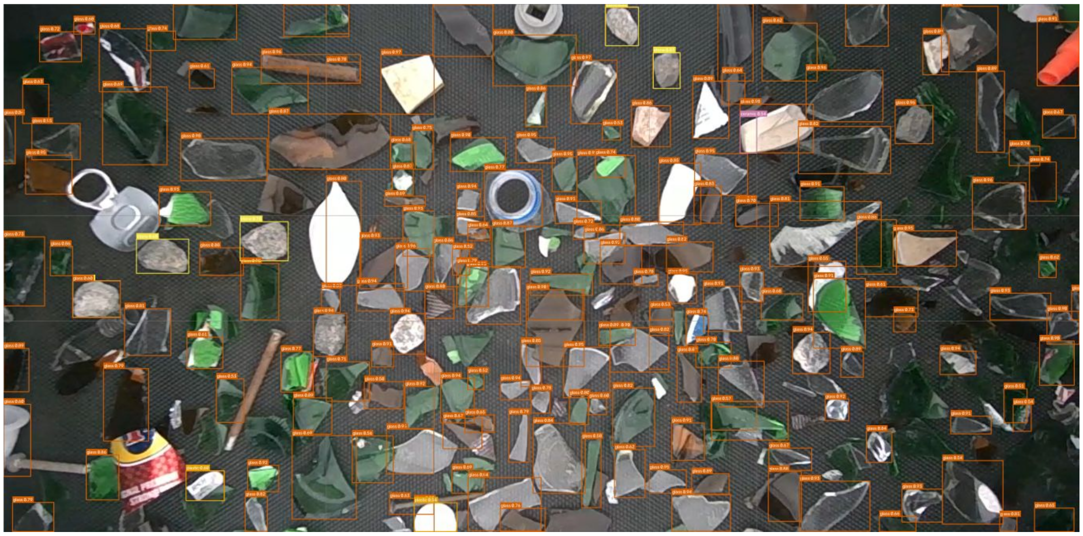


Prediction

Plain inference (no tiling):



SAHI inference (6-tile):



2. YOLO26s

Model Explanation

Same YOLO26 architecture family as exp006, but trained as a narrow specialist: a single output class (glass cullet) on a dataset roughly a third the size of the main six-class set. Glass-cullet fragments have no consistent silhouette the way a whole bottle or can does, which - combined with the smaller dataset - explains its lower headline numbers versus the six-class models.

Dataset

Dataset is a separate, narrower capture set of glass-cullet fragments only - 2,350 training images and 587 validation images, single class.

Training Parameters

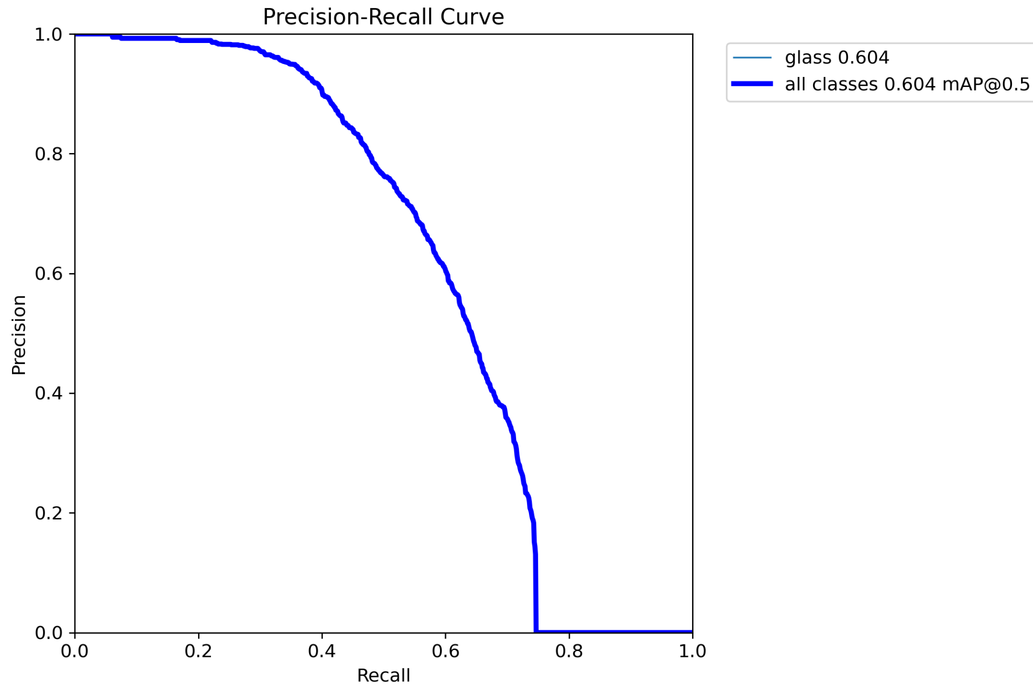
Base model	YOLO26s
Classes	1 - glass (cullet fragments only)
Training images	2,350
Validation images	587
Batch size	8
Training Image Input size	640px
Epochs trained	120
Training time	1h 23m (120 epochs)
Precision	0.694
Recall	0.553
mAP50	0.602
mAP50-95	0.370

Average model Inference time is : 6.03 ms (165.7 FPS)

Average end to end latency : 79.87 ms

(Measured on an RTX 5080, 3622x1779 test frame, 5 warmup + 30 timed passes. End-to-end includes reading the frame and drawing the annotated output, not just the model's forward pass.)

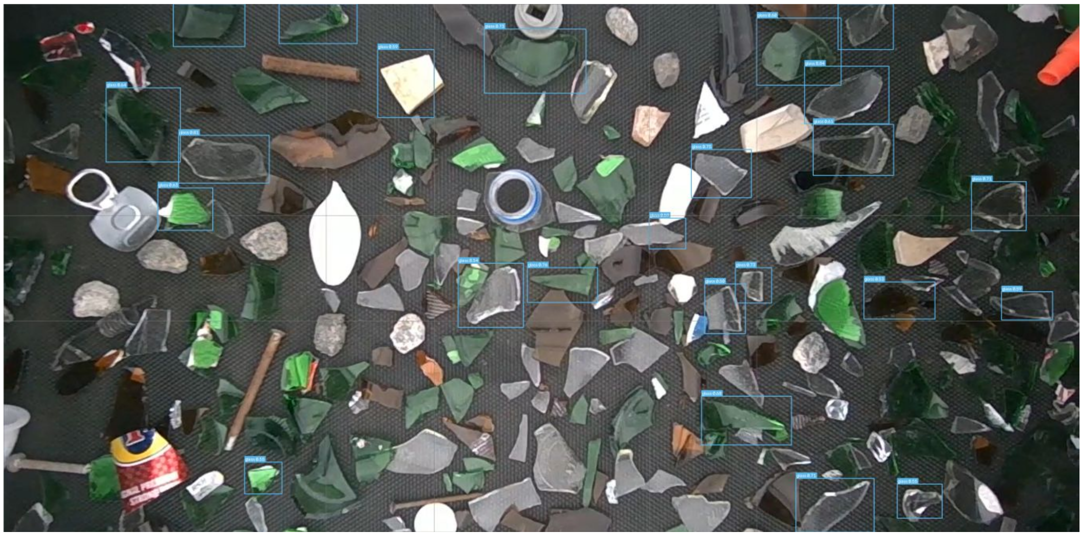
Model curve



Prediction

Plain inference (no tiling):

SAHI inference (6-tile):



3. YOLO26L + P2

Model Explanation

The only architectural change from a stock YOLO26L is an added P2 (160x160) detection head alongside the stock P3/P4/P5 heads, giving the network a detection path with less downsampling - better suited to small or distant objects on the conveyor. Every other hyperparameter is left at Ultralytics defaults, isolating the P2 head's own effect on accuracy. This is the best-performing model in the project on every headline metric.

Dataset

Same main dataset as YOLOv8L above - 7,389 training images and 1,847 validation images, identical split for a fair comparison.

Training Parameters

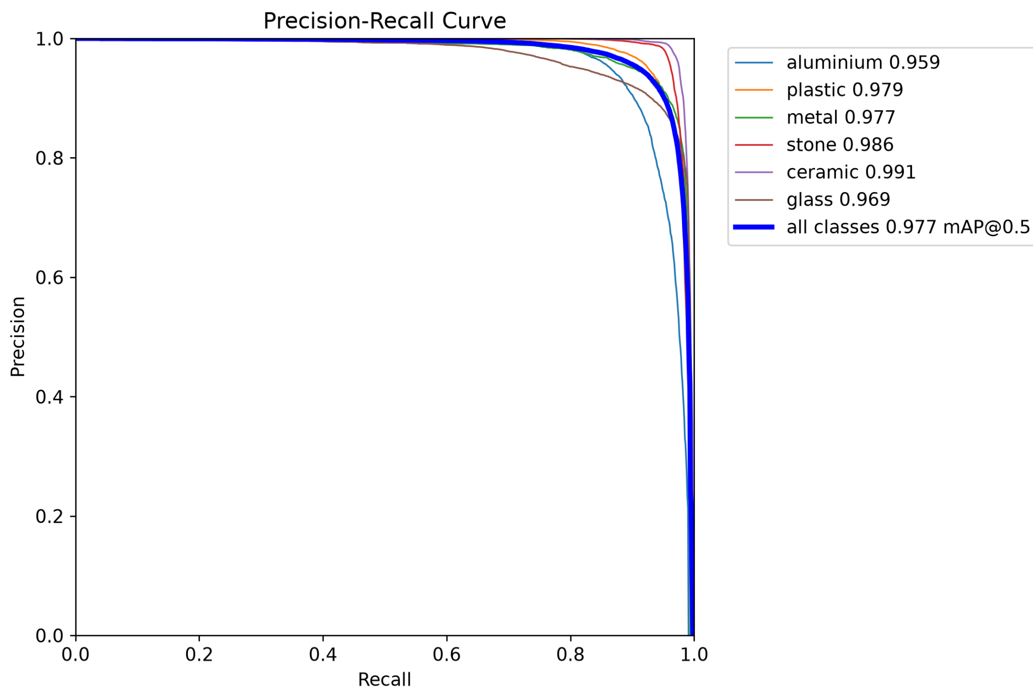
Base model	YOLO26L + P2
Classes	6 - aluminium, plastic, metal, stone, ceramic, glass
Training images	7,389
Validation images	1,847
Batch size	16
Training Image Input size	640px
Epochs trained	100
Training time	41h 17m (100 epochs)
Precision	0.947
Recall	0.930
mAP50	0.977
mAP50-95	0.951

Average model Inference time is : 11.6 ms (86.2 FPS)

Average end to end latency : 91.74 ms

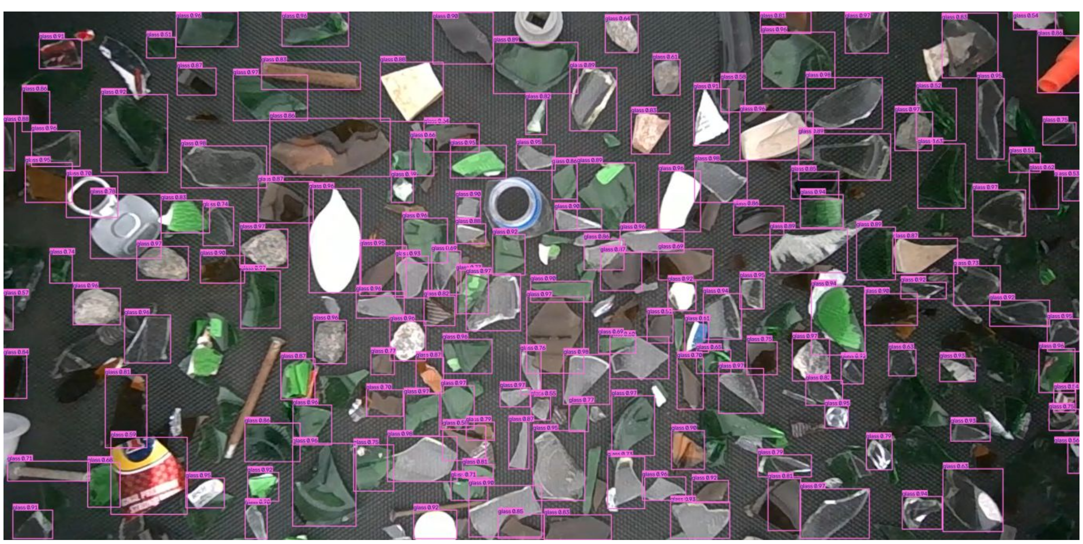
(Measured on an RTX 5080, 3622x1779 test frame, 5 warmup + 30 timed passes. End-to-end includes reading the frame and drawing the annotated output, not just the model's forward pass.)

Model curve

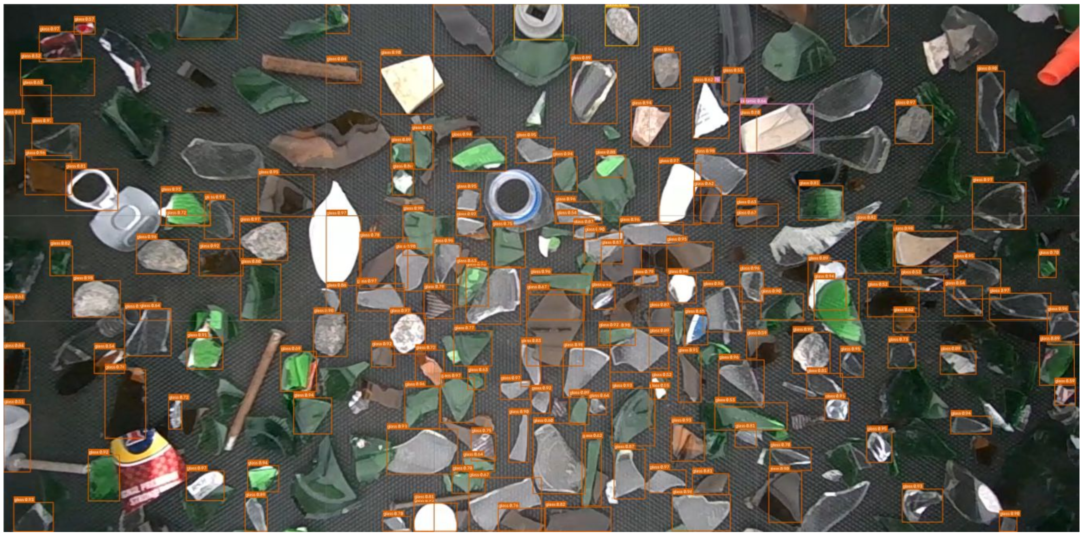


Prediction

Plain inference (no tiling):



SAHI inference (6-tile):



4. ResNet-50

Model Explanation

A standard ResNet-50 backbone, ImageNet-pretrained, with its final layer replaced by a 6-way linear classifier. Not a detector: it is the second stage of a two-model pipeline where YOLO detects and localizes an object first, then this classifier looks only at that tightly cropped region to assign the final material label, using heavy color/lighting augmentation for robustness to glare. It cannot localize objects in a raw multi-object frame on its own.

Dataset

Dataset is not separately captured - it is built by cropping every labeled YOLO box out of the main dataset's own images (10% padding added, boxes under 10px skipped), giving 294,959 training crops and 73,448 validation crops with zero new manual labeling.

Training Parameters

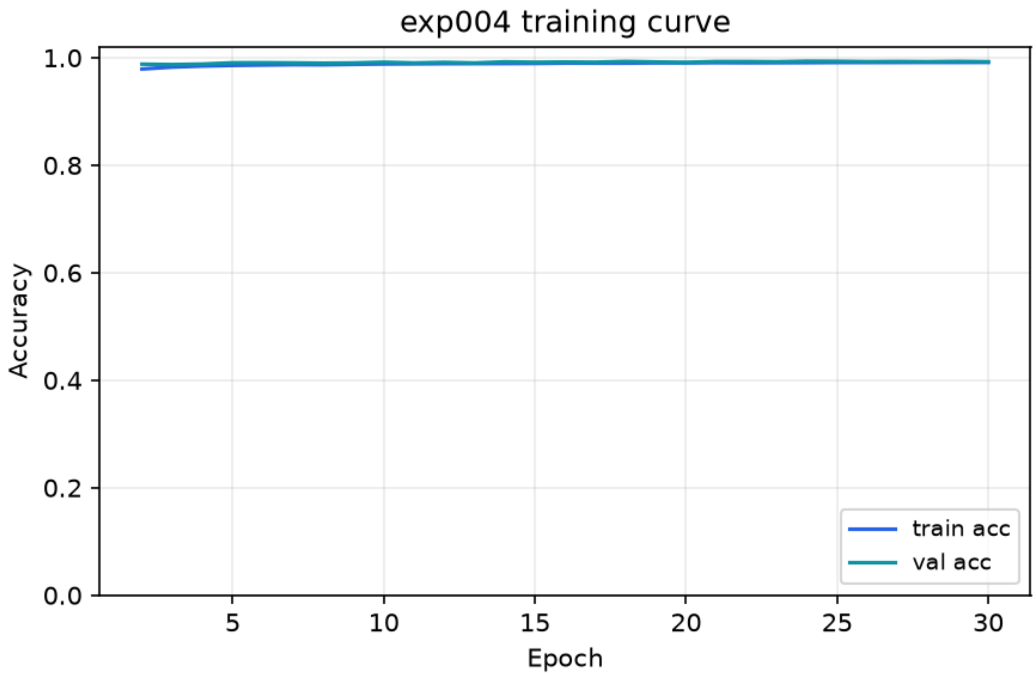
Base model	ResNet-50
Classes	6 - aluminium, plastic, metal, stone, ceramic, glass
Training crops	294,959 crops
Validation crops	73,448 crops
Batch size	64
Training Image Input size	224px
Epochs trained	29
Training time	not logged for this run (29 epochs to best checkpoint)
Accuracy	99.37%

Average model Inference time is : 3.33 ms (300.2 FPS)

Average end to end latency : 77.63 ms

(Measured on an RTX 5080, 3622x1779 test frame, 5 warmup + 30 timed passes. End-to-end includes reading the frame and drawing the annotated output, not just the model's forward pass.)

Model curve



Prediction

Plain inference (no tiling):

SAHI inference (6-tile):

5. RT-DETR-L

Model Explanation

RT-DETR (Baidu, 2023) is a transformer-based detector with no anchor boxes and no NMS post-processing - a CNN backbone feeds a transformer encoder-decoder that predicts a fixed set of object queries directly. Trained via Ultralytics' own RTDETR class, the same .train() API as every YOLO model here, making it a fair architecture-only comparison. Training completed all 100 epochs, but the process was interrupted during Ultralytics' final validation pass, so no confusion matrix or PR-curve image exists for this run - the chart below is a live per-epoch training curve instead.

Dataset

Same main dataset and identical validation split as YOLOv8L and YOLO26L+P2 above - 7,389 training images and 1,847 validation images.

Training Parameters

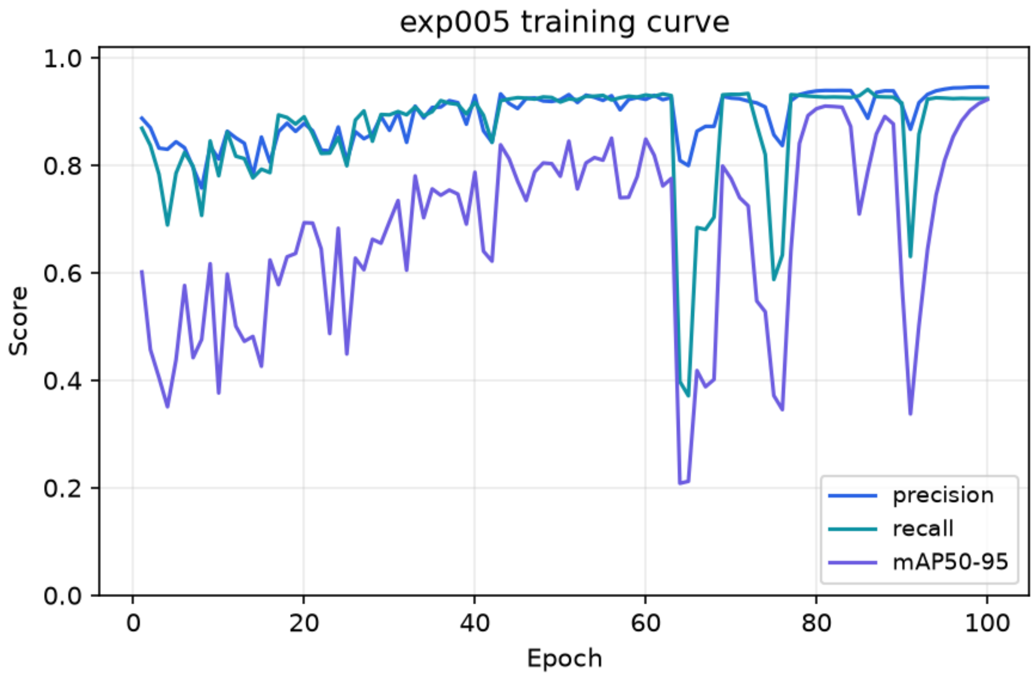
Base model	RT-DETR-L
Classes	6 - aluminium, plastic, metal, stone, ceramic, glass
Training images	7,389
Validation images	1,847
Batch size	16
Training Image Input size	640px
Epochs trained	100
Training time	4h 48m (100 epochs)
Precision	0.946
Recall	0.924
mAP50	0.975
mAP50-95	0.922

Average model Inference time is : 13.93 ms (71.8 FPS)

Average end to end latency : 96.87 ms

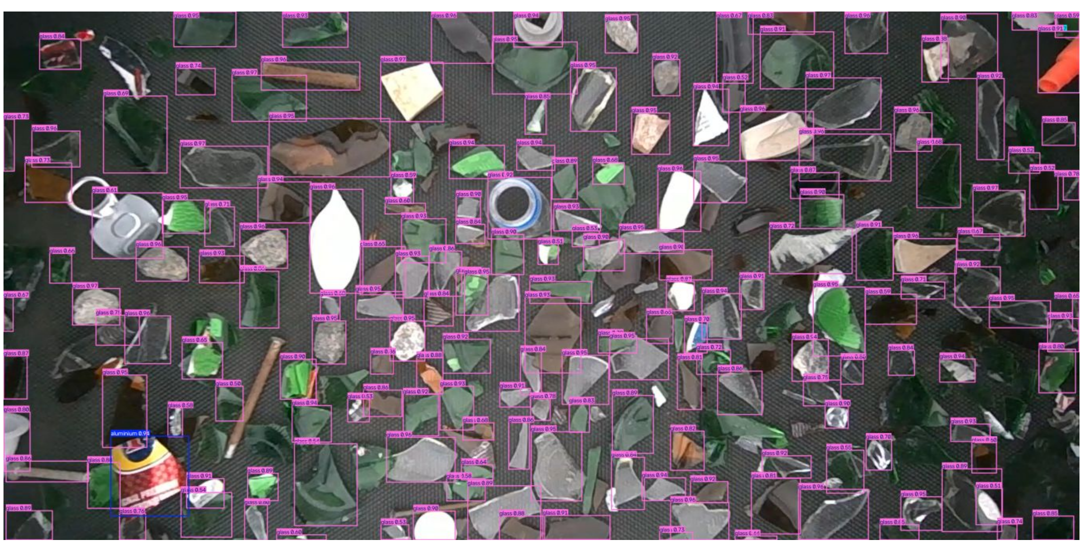
(Measured on an RTX 5080, 3622x1779 test frame, 5 warmup + 30 timed passes. End-to-end includes reading the frame and drawing the annotated output, not just the model's forward pass.)

Model curve

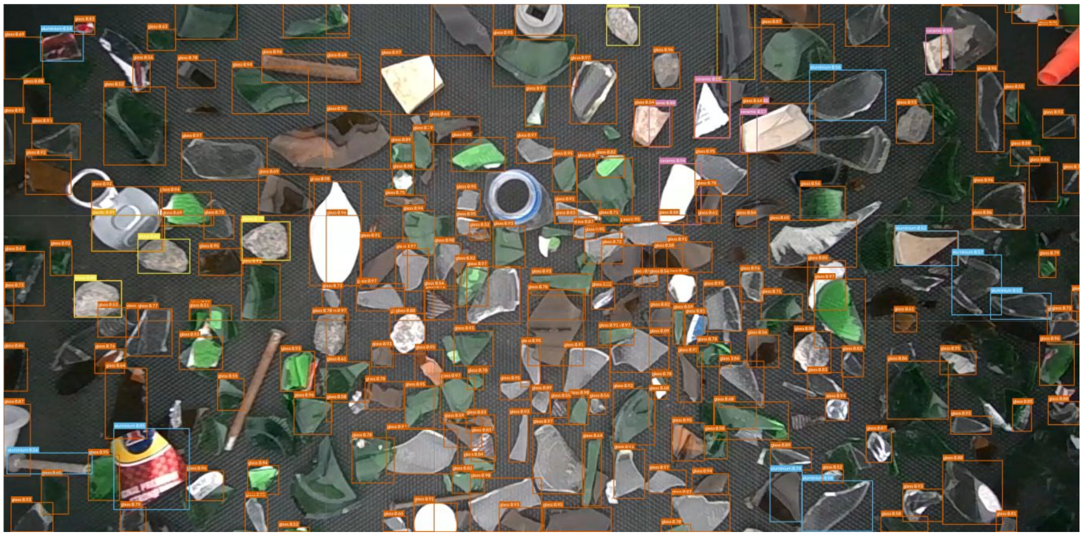


Prediction

Plain inference (no tiling):



SAHI inference (6-tile):



6. YOLO26s (fine-tuned)

Model Explanation

Same YOLO26s architecture as exp002, trained in two stages instead of one: stage 1 base-trains on a separate smaller dataset, then stage 2 fine-tunes the resulting weights on this project's main six-class dataset. It lands between YOLOv8L and YOLO26L+P2 on the final held-out set despite being a much smaller, faster network - evidence that the base-training stage transfers useful structure before the model ever sees this project's own data.

Dataset

Trained in two stages: base-trained on its own 1,631/408 dataset, then fine-tuned on the main 7,389/1,847 dataset - the numbers below are the final fine-tuned model.

Training Parameters

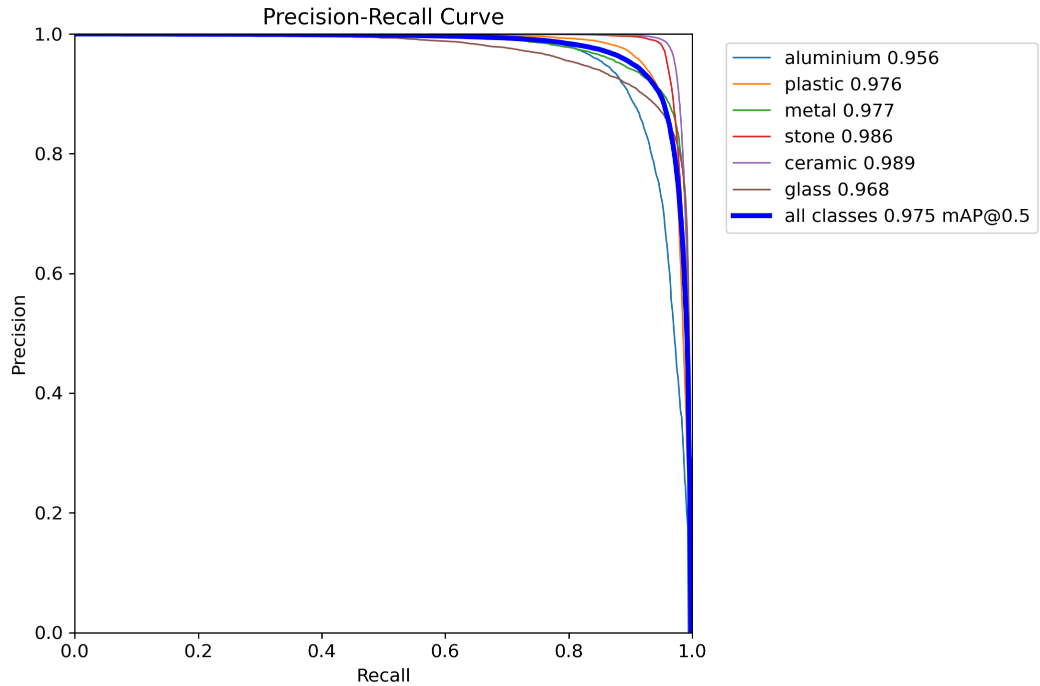
Base model	YOLO26s (fine-tuned)
Classes	6 - aluminium, plastic, metal, stone, ceramic, glass
Training images	1,631 + 7,389
Validation images	408 / 1,847
Batch size	8
Training Image Input size	640px
Epochs trained	100
Training time	29m (fine-tune stage, 100 epochs) + a separate base-training stage
Precision	0.944
Recall	0.927
mAP50	0.975
mAP50-95	0.943

Average model Inference time is : 6.7 ms (149.2 FPS)

Average end to end latency : 85.24 ms

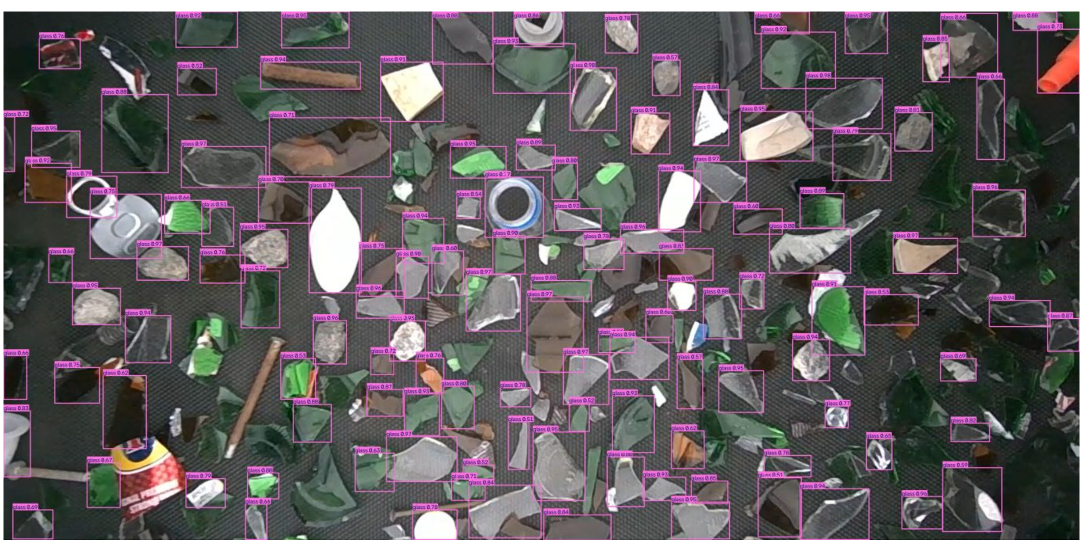
(Measured on an RTX 5080, 3622x1779 test frame, 5 warmup + 30 timed passes. End-to-end includes reading the frame and drawing the annotated output, not just the model's forward pass.)

Model curve



Prediction

Plain inference (no tiling):



SAHI inference (6-tile):